

THE ULTIMATE ORF309 MIDTERM 2 STUDY GUIDE

Everything You Need to Ace This Exam

One Guide. All Topics. Step-by-Step.

From Zero to Hero in 3-4 Days

What's Inside:

- Beginner-friendly explanations with real-world examples
- Every formula you need (color-coded by importance)
- 15+ fully worked examples from past exams
- Pattern recognition guide (when to use what technique)
- Common mistakes and how to avoid them
- Step-by-step problem-solving strategies

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START HERE: Quick Navigation

First Time Learning? Read in order: Section 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4

Need Quick Review? Go straight to “Formula Sheets” (Section 8)

Doing Practice Problems? Use “Pattern Recognition Guide” (Section 7)

Last-Minute Cramming? Focus on Sections 1, 2, and 8 only

The Big 3 Topics (Guaranteed on Exam)

1. **Poisson Processes** (Section 1) - 40% of exam, appears in 100% of exams
 - Master thinning, competing processes, conditional expectations
2. **Random Walks** (Section 2) - 35% of exam, appears in 100% of exams
 - Master first-step analysis for hitting probabilities and expected times
3. **Joint/Conditional Densities** (Section 3) - 25% of exam, appears in 86% of exams
 - Master law of total expectation and building joint PDFs

How to Use This Guide

Day 1: Read Section 1 (Poisson), do all examples, try 2 practice problems

Day 2: Read Section 2 (Random Walks), do all examples, try 2 practice problems

Day 3: Read Section 3 (Joint Densities), review Sections 4-5, do full practice exam

Day 4: Review Section 8 (formulas), do another practice exam, make cheat sheet

1 POISSON PROCESSES - The Most Important Topic

CRITICAL - MASTER THIS FIRST

Why This Matters: Appears in 7/7 exams (100%). Average weight: 40% of total points.
Master thinning and conditional expectations and you'll solve half the exam!

1.1 What is a Poisson Process?

1.1.1 The Big Idea (In Plain English)

Imagine sitting at Starbucks counting customers. Sometimes 5 people arrive in 2 minutes, sometimes nobody for 10 minutes. Arrivals seem random, but there's an *average rate*.

A **Poisson process** models these random arrivals:

- Events happen randomly over time
- There's an average rate λ (like "3 customers per minute")
- Each event is independent of others

1.1.2 The Basic Math

If events arrive at rate λ (per unit time):

KEY FORMULAS

Number of arrivals in time t : $N(t) \sim \text{Poisson}(\lambda t)$

Probability of exactly k arrivals:

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

Expected number of arrivals: $E[N(t)] = \lambda t$

Variance: $\text{Var}(N(t)) = \lambda t$

WORKED EXAMPLE

Problem: Emails arrive at your inbox at rate $\lambda = 10$ per hour. How many emails do you expect in 3 hours?

Solution:

$$E[N(3)] = \lambda t = 10 \times 3 = \boxed{30 \text{ emails}}$$

What's the probability of exactly 25 emails in 3 hours?

$$\begin{aligned} P(N(3) = 25) &= \frac{(10 \cdot 3)^{25} e^{-10 \cdot 3}}{25!} \\ &= \frac{30^{25} e^{-30}}{25!} \approx 0.0387 \end{aligned}$$

1.2 THINNING - The Most Tested Concept

CRITICAL - MASTER THIS FIRST

Appears in 6/7 exams! If you master ONE thing, make it thinning.

1.2.1 The Setup

Events arrive at rate λ . Each event is independently classified as:

- "Type A" with probability p
- "Type B" with probability $1 - p$

Question: What's the rate of Type A events?

1.2.2 The Answer (Thinning Property)

KEY FORMULAS

THINNING PROPERTY:

If arrivals are $\text{Poisson}(\lambda)$ and each is Type A with probability p :

$$\text{Type A arrivals are } \text{Poisson}(\lambda p)$$

You “thin” the arrival rate by multiplying by the probability!

1.2.3 Intuition

If 10 customers arrive per minute and 30% order coffee:

$$\text{Coffee orders} = 10 \times 0.3 = 3 \text{ per minute}$$

WORKED EXAMPLE

Problem (Exam Style): At a coffee shop, customers arrive at rate 2 per minute. Each customer:

- Orders coffee with probability 0.4
- Orders tea with probability 0.4
- Orders food with probability 0.2

(a) What’s the rate of coffee orders?

Solution (a): Use thinning:

$$\lambda_{\text{coffee}} = 2 \times 0.4 = \boxed{0.8 \text{ per minute}}$$

(b) On average, how many coffee orders in 10 minutes?

Solution (b):

$$E[\text{coffee in 10 min}] = \lambda_{\text{coffee}} \times t = 0.8 \times 10 = \boxed{8 \text{ coffee orders}}$$

(c) What’s the probability that exactly 4 coffee orders happen before the first food order?

Solution (c): Think of it as a race:

- Coffee orders: $\text{Poisson}(0.8/\text{min})$
- Food orders: $\text{Poisson}(0.4/\text{min})$

For exactly 4 coffee before first food, we want the sequence: Coffee, Coffee, Coffee, Coffee, Food.

Key insight: Tea orders don’t matter! Among coffee+food orders only, each one is independently:

- Coffee with probability: $\frac{0.8}{0.8+0.4} = \frac{2}{3}$
- Food with probability: $\frac{0.4}{0.8+0.4} = \frac{1}{3}$

So we need the specific sequence C-C-C-C-F:

$$P(4 \text{ coffee before 1 food}) = \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right) = \frac{16}{81 \times 3} = \boxed{\frac{16}{243}}$$

Why doesn’t tea matter? Tea customers can arrive in between, but we only care about the coffee vs food “race.” It’s like watching only red and blue cars pass by and asking “what’s the probability I see 4 red before 1 blue?” - green cars (tea) passing by don’t affect this.

1.3 Competing Processes

1.3.1 The Setup

Two independent Poisson processes:

- Process A at rate λ_A
- Process B at rate λ_B

Question: Which happens first?

1.3.2 The Formulas

KEY FORMULAS

COMPETING PROCESSES:

Probability A happens before B:

$$P(\text{A before B}) = \frac{\lambda_A}{\lambda_A + \lambda_B}$$

Expected time until first event (A or B):

$$E[T] = \frac{1}{\lambda_A + \lambda_B}$$

The first event to occur: $\text{Poisson}(\lambda_A + \lambda_B)$

WORKED EXAMPLE

Problem: You're waiting for:

- Uber: arrives at rate $\lambda_U = 6$ per hour
- Bus: arrives at rate $\lambda_B = 4$ per hour

(a) What's the probability Uber arrives first?

Solution (a):

$$P(\text{Uber first}) = \frac{6}{6 + 4} = \frac{6}{10} = \boxed{0.6 = 60\%}$$

(b) How long do you expect to wait?

Solution (b):

$$E[T] = \frac{1}{\lambda_U + \lambda_B} = \frac{1}{6 + 4} = \frac{1}{10} \text{ hour} = \boxed{6 \text{ minutes}}$$

1.4 Conditional Expectations with Poisson

CRITICAL - MASTER THIS FIRST

Appears in 5/7 exams! This is almost as important as thinning.

1.4.1 The Setup

Events arrive at rate λ . You're told that exactly n events occurred.

Question: What's the expected time?

1.4.2 The Formula

KEY FORMULAS

CONDITIONAL EXPECTATION:

Given that exactly n arrivals occurred by time T :

$$\boxed{E[T \mid N(T) = n] = \frac{n}{\lambda}}$$

Intuition: If n events happened at rate λ , it must have taken n/λ time.

WORKED EXAMPLE

Problem: Customers arrive at rate $\lambda = 2$ per minute. Given that exactly 10 customers arrived, what's the expected time?

Solution:

$$E[T \mid N(T) = 10] = \frac{10}{2} = \boxed{5 \text{ minutes}}$$

Problem (Harder): At a taxi stop, taxis arrive at rate 1/min, passengers at rate 2/min. Given that by the time the first taxi arrives, exactly 10 passengers are waiting, what's the expected time?

Solution:

- First taxi arrival time: $T \sim \text{Exp}(1)$
- Given 10 passengers arrived: $E[T \mid N_{\text{pass}}(T) = 10] = \frac{10}{2} = \boxed{5 \text{ minutes}}$

1.5 Poisson Quick Reference

KEY FORMULAS**POISSON PROCESSES - FORMULA SHEET**

Basic Facts:

$$\begin{aligned} E[N(t)] &= \lambda t \\ \text{Var}(N(t)) &= \lambda t \\ P(N(t) = k) &= \frac{(\lambda t)^k e^{-\lambda t}}{k!} \end{aligned}$$

Thinning: Type A prob $p \rightarrow$ Type A is Poisson(λp)

Competing: $P(\text{A before B}) = \frac{\lambda_A}{\lambda_A + \lambda_B}$

Conditional: $E[T \mid N(T) = n] = \frac{n}{\lambda}$

COMMON MISTAKE - AVOID THIS!

Common Mistakes:

1. Forgetting to use thinning when events split into types
2. Using λ instead of λt in Poisson formula
3. Not recognizing conditional expectation problems

2 RANDOM WALKS & FIRST-STEP ANALYSIS

CRITICAL - MASTER THIS FIRST

Why This Matters: Appears in **7/7 exams (100%)**. Average weight: **35% of total points**. Master first-step analysis and you've conquered another third of the exam!

2.1 What is a Random Walk?

2.1.1 The Big Idea (In Plain English)

Imagine a gambling game:

- You start with \$50
 - Each round: flip a coin
 - Heads: win \$1, Tails: lose \$1
 - Game ends at \$0 (bankrupt) or \$100 (rich)
- Your money randomly walks up and down. This is a **random walk**!

2.1.2 Formal Setup

- **States:** Positions $a, a+1, a+2, \dots, b-1, b$
- **Barriers:** a (lower) and b (upper) where game ends
- **Starting position:** $S_0 = i$ (somewhere between a and b)
- **Transitions:** Each step, move up/down/stay with given probabilities

2.2 The Two Main Questions

Every random walk problem asks one (or both) of:

1. **Hitting Probability:** What's $P(\text{reach } b \text{ before } a)$?
2. **Expected Hitting Time:** How long until we reach a or b ?

2.3 FIRST-STEP ANALYSIS - The Key Technique

CRITICAL - MASTER THIS FIRST

This technique appears in **100% of exams**. Master it and random walk problems become easy!

2.3.1 The Core Idea

To figure out what happens from position i :

1. Think about what happens on the *first step*
2. Use what you know about where you'll be next
3. Write a recursive equation

It's like saying: "My chance of winning from \$50 depends on whether I go to \$51 or \$49 next, and my chances from those positions."

2.4 First-Step Analysis: HITTING PROBABILITY

2.4.1 Setup

Let $p(i)$ = probability of reaching b before a , starting from position i .

2.4.2 The Recursive Formula

KEY FORMULAS

FIRST-STEP ANALYSIS FOR HITTING PROBABILITY:

Recursive formula:

$$p(i) = p_{\uparrow} \cdot p(i+1) + p_{\downarrow} \cdot p(i-1) + p_{\leftrightarrow} \cdot p(i)$$

where $p_{\uparrow}$ = prob of moving up, $p_{\downarrow}$ = prob of moving down, $p_{\leftrightarrow}$ = prob of staying

Boundary conditions:

$$p(a) = 0 \quad \text{and} \quad p(b) = 1$$

Solve the system of equations for all intermediate states!

2.4.3 Special Case: Symmetric Random Walk

When $p_{\uparrow} = p_{\downarrow} = 1/2$ (fair coin flip):

KEY FORMULAS

SYMMETRIC WALK SHORTCUT:

$$p(i) = \frac{i - a}{b - a}$$

Your hitting probability is just your relative position between the barriers!

WORKED EXAMPLE

Problem: You play a game with positions 0, 1, 2, 3, 4. Each round:

- Move up with probability $1/2$
- Move down with probability $1/2$

Start at position 2. Win at 4, lose at 0.

What's the probability of winning?

Solution (Method 1 - Shortcut):

This is symmetric, so:

$$p(2) = \frac{2-0}{4-0} = \frac{2}{4} = \boxed{0.5 = 50\%}$$

Solution (Method 2 - First-Step Analysis):

Boundaries: $p(0) = 0$, $p(4) = 1$

From position 1:

$$p(1) = \frac{1}{2}p(2) + \frac{1}{2}p(0) = \frac{1}{2}p(2)$$

From position 2:

$$p(2) = \frac{1}{2}p(3) + \frac{1}{2}p(1)$$

From position 3:

$$p(3) = \frac{1}{2}p(4) + \frac{1}{2}p(2) = \frac{1}{2} + \frac{1}{2}p(2)$$

Substitute $p(1) = \frac{1}{2}p(2)$ and $p(3) = \frac{1}{2} + \frac{1}{2}p(2)$ into $p(2)$ equation:

$$p(2) = \frac{1}{2} \left(\frac{1}{2} + \frac{1}{2}p(2) \right) + \frac{1}{2} \left(\frac{1}{2}p(2) \right)$$

$$p(2) = \frac{1}{4} + \frac{1}{4}p(2) + \frac{1}{4}p(2)$$

$$p(2) = \frac{1}{4} + \frac{1}{2}p(2)$$

$$\frac{1}{2}p(2) = \frac{1}{4}$$

$$p(2) = \boxed{0.5}$$

2.5 First-Step Analysis: EXPECTED TIME

CRITICAL - MASTER THIS FIRST

Most Common Mistake: Forgetting the “+1” in this formula! Don't be that person!

2.5.1 Setup

Let $\mu(i)$ = expected number of steps to reach a or b , starting from position i .

2.5.2 The Recursive Formula

KEY FORMULAS

FIRST-STEP ANALYSIS FOR EXPECTED TIME:

Recursive formula:

$$\mu(i) = 1 + p_{\uparrow} \cdot \mu(i+1) + p_{\downarrow} \cdot \mu(i-1) + p_{\leftrightarrow} \cdot \mu(i)$$

DON'T FORGET THE “+1”! (Counts the current step)

Boundary conditions:

$$\mu(a) = 0 \quad \text{and} \quad \mu(b) = 0$$

(Game is over at the barriers!)

2.5.3 Special Case: Symmetric Random Walk

When $p_{\uparrow} = p_{\downarrow} = 1/2$:

KEY FORMULAS

SYMMETRIC WALK SHORTCUT:

$$\mu(i) = (i - a)(b - i)$$

WORKED EXAMPLE

Problem: Same game as before (positions 0–4, symmetric). Start at 2.

How many steps on average until the game ends?

Solution (Method 1 - Shortcut):

$$\mu(2) = (2 - 0)(4 - 2) = 2 \times 2 = \boxed{4 \text{ steps}}$$

Solution (Method 2 - First-Step Analysis):

Boundaries: $\mu(0) = 0$, $\mu(4) = 0$

From position 1:

$$\mu(1) = 1 + \frac{1}{2}\mu(2) + \frac{1}{2}\mu(0) = 1 + \frac{1}{2}\mu(2)$$

From position 2:

$$\mu(2) = 1 + \frac{1}{2}\mu(3) + \frac{1}{2}\mu(1)$$

From position 3:

$$\mu(3) = 1 + \frac{1}{2}\mu(4) + \frac{1}{2}\mu(2) = 1 + \frac{1}{2}\mu(2)$$

Notice $\mu(1) = \mu(3) = 1 + \frac{1}{2}\mu(2)$ by symmetry.

Substitute into $\mu(2)$ equation:

$$\mu(2) = 1 + \frac{1}{2}(1 + \frac{1}{2}\mu(2)) + \frac{1}{2}(1 + \frac{1}{2}\mu(2))$$

$$\mu(2) = 1 + \frac{1}{2} + \frac{1}{4}\mu(2) + \frac{1}{2} + \frac{1}{4}\mu(2)$$

$$\mu(2) = 2 + \frac{1}{2}\mu(2)$$

$$\frac{1}{2}\mu(2) = 2$$

$$\mu(2) = \boxed{4}$$

2.6 Biased Random Walks

2.6.1 When Probabilities Are Not Equal

Sometimes $P(\text{up}) \neq P(\text{down})$. This is a **biased random walk**.

Approach: Use first-step analysis and solve the system of equations (no shortcuts!).

WORKED EXAMPLE

Problem (Exam-Style): Positions 0, 1, 2, 3. Each step:

- Move up with probability $2/3$
- Move down with probability $1/3$

Start at position 1. What's $p(1)$?

Solution:

Boundaries: $p(0) = 0$, $p(3) = 1$

From position 1:

$$p(1) = \frac{2}{3}p(2) + \frac{1}{3}p(0) = \frac{2}{3}p(2)$$

From position 2:

$$p(2) = \frac{2}{3}p(3) + \frac{1}{3}p(1) = \frac{2}{3} + \frac{1}{3}p(1)$$

Substitute $p(1) = \frac{2}{3}p(2)$ into second equation:

$$\begin{aligned} p(2) &= \frac{2}{3} + \frac{1}{3} \cdot \frac{2}{3}p(2) \\ p(2) &= \frac{2}{3} + \frac{2}{9}p(2) \\ \frac{7}{9}p(2) &= \frac{2}{3} \\ p(2) &= \frac{2}{3} \cdot \frac{9}{7} = \frac{6}{7} \end{aligned}$$

Therefore:

$$p(1) = \frac{2}{3} \cdot \frac{6}{7} = \boxed{\frac{4}{7}}$$

2.7 Random Walks Quick Reference

KEY FORMULAS

RANDOM WALKS - FORMULA SHEET

Hitting Probability $p(i)$:

$$p(i) = p_{\uparrow}p(i+1) + p_{\downarrow}p(i-1) + p_{\leftrightarrow}p(i)$$

Boundaries: $p(a) = 0$, $p(b) = 1$

Expected Time $\mu(i)$:

$$\mu(i) = 1 + p_{\uparrow}\mu(i+1) + p_{\downarrow}\mu(i-1) + p_{\leftrightarrow}\mu(i)$$

Boundaries: $\mu(a) = 0$, $\mu(b) = 0$

Symmetric Walk Shortcuts ($p_{\uparrow} = p_{\downarrow} = 1/2$):

$$\begin{aligned} p(i) &= \frac{i-a}{b-a} \\ \mu(i) &= (i-a)(b-i) \end{aligned}$$

COMMON MISTAKE - AVOID THIS!**Common Mistakes:**

1. **Forgetting “+1”** in $\mu(i)$ formula (MOST COMMON!)
2. Wrong boundary conditions (μ should be 0 at BOTH barriers)
3. Not setting up all equations (need one for each intermediate state)
4. Arithmetic errors when solving systems

PROBLEM-SOLVING STRATEGY**HOW TO SOLVE RANDOM WALK PROBLEMS:**

- Step 1:** Draw a diagram with all states
- Step 2:** Identify what you're solving for ($p(i)$ or $\mu(i)$?)
- Step 3:** Check if symmetric $\rightarrow$ use shortcut if possible
- Step 4:** Write boundary conditions
- Step 5:** Write first-step equation for each intermediate state
- Step 6:** Solve the system (usually 2-3 equations, 2-3 unknowns)
- Step 7:** Check answer makes intuitive sense

3 JOINT & CONDITIONAL DENSITIES

CRITICAL - MASTER THIS FIRST

Why This Matters: Appears in 6/7 exams (86%). Average weight: 25% of total points.

3.1 What Are We Dealing With?

3.1.1 The Big Idea

Sometimes we have **two related random variables**:

- X = total restaurant bill
- Y = amount you pay (depends on X !)
 - We need to understand how they relate using **joint** and **conditional** densities.

3.2 Key Definitions

KEY FORMULAS

JOINT PDF: $f_{X,Y}(x,y)$ describes both variables together

MARGINAL PDF: Distribution of one variable alone

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

CONDITIONAL PDF: Distribution of one given the other

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \quad f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

BUILDING JOINT FROM CONDITIONAL:

$$f_{X,Y}(x,y) = f_{Y|X}(y|x) \cdot f_X(x) = f_{X|Y}(x|y) \cdot f_Y(y)$$

3.3 Law of Total Expectation

CRITICAL - MASTER THIS FIRST

This is one of the most powerful tools in probability. Master it!

KEY FORMULAS

LAW OF TOTAL EXPECTATION (Tower Property):

$$E[Y] = E[E[Y|X]]$$

In integral form:

$$E[Y] = \int_{-\infty}^{\infty} E[Y|X=x] \cdot f_X(x) dx$$

When to use: When Y depends on X and you know the conditional expectation.

3.4 The $Y = UX$ Transformation (Super Common!)

This appears in many exam problems:

- X = some random variable (e.g., bill amount, loss amount)
- $U \sim \text{Uniform}[0, 1]$, independent of X
- $Y = UX$ = fraction of X

KEY FORMULAS

KEY FACTS FOR $Y = UX$:

Given $X = x$:

- $Y = Ux$ is uniform on $[0, x]$
- $f_{Y|X}(y|x) = \frac{1}{x}$ for $0 < y < x$
- $E[Y|X = x] = \frac{x}{2}$

Joint PDF:

$$f_{X,Y}(x, y) = f_{Y|X}(y|x) \cdot f_X(x) = \frac{1}{x} f_X(x) \quad \text{for } 0 < y < x$$

Always specify the range!

WORKED EXAMPLE

Problem (Exam-Style):

- X = total loss (in \$1000s), with $P(X > x) = 1/x^2$ for $x \geq 1$
- $U \sim \text{Uniform}[0, 1]$, independent of X
- $Y = UX$ = amount you pay out-of-pocket

(a) Find $E[Y]$ (expected payment).

Solution (a): Use law of total expectation!

Step 1: Find $E[Y|X = x]$

$$E[Y|X = x] = E[UX|X = x] = xE[U] = x \cdot \frac{1}{2} = \frac{x}{2}$$

Step 2: Find $f_X(x)$

$$P(X > x) = 1/x^2 \Rightarrow F_X(x) = 1 - 1/x^2$$

$$f_X(x) = \frac{d}{dx} F_X(x) = \frac{2}{x^3} \quad \text{for } x \geq 1$$

Step 3: Apply law of total expectation

$$E[Y] = E[E[Y|X]] = E\left[\frac{X}{2}\right] = \frac{1}{2}E[X]$$

$$\begin{aligned} E[X] &= \int_1^\infty x \cdot \frac{2}{x^3} dx = \int_1^\infty \frac{2}{x^2} dx \\ &= \left[-\frac{2}{x}\right]_1^\infty = 0 - (-2) = 2 \end{aligned}$$

Therefore:

$$E[Y] = \frac{1}{2} \cdot 2 = \boxed{\$1000}$$

(b) Find joint PDF $f_{X,Y}(x, y)$.

Solution (b):

$$f_{X,Y}(x, y) = f_{Y|X}(y|x) \cdot f_X(x) = \frac{1}{x} \cdot \frac{2}{x^3} = \boxed{\frac{2}{x^4}} \quad \text{for } 0 < y < x, x \geq 1$$

CRITICAL: Always specify the range!

(c) Given you paid exactly \$1000 ($Y = 1$), what's $P(X > 2)$?

Solution (c): Need conditional probability $P(X > 2|Y = 1)$.

First find $f_Y(1)$:

$$\begin{aligned} f_Y(1) &= \int_1^\infty f_{X,Y}(x, 1) dx \\ &= \int_1^\infty \frac{2}{x^4} dx = \left[-\frac{2}{3x^3}\right]_1^\infty = \frac{2}{3} \end{aligned}$$

Then:

$$\begin{aligned} P(X > 2|Y = 1) &= \frac{\int_2^\infty f_{X,Y}(x, 1) dx}{f_Y(1)} \\ &= \frac{\int_2^\infty \frac{2}{x^4} dx}{2/3} = \frac{[-2/(3x^3)]_2^\infty}{2/3} \\ &= \frac{2/(3 \cdot 8)}{2/3} = \frac{1/12}{2/3} = \boxed{\frac{1}{8}} \end{aligned}$$

3.5 Joint Densities Quick Reference

KEY FORMULAS

JOINT/CONDITIONAL DENSITIES - FORMULA SHEET

Relationships:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_{X,Y}(x,y) = f_{Y|X}(y|x) \cdot f_X(x)$$

$$f_X(x) = \int f_{X,Y}(x,y) dy$$

Law of Total Expectation:

$$E[Y] = E[E[Y|X]]$$

For $Y = UX$ where $U \sim \text{Unif}[0, 1]$:

$$f_{Y|X}(y|x) = \frac{1}{x} \quad (0 < y < x)$$

$$E[Y|X = x] = \frac{x}{2}$$

$$f_{X,Y}(x,y) = \frac{f_X(x)}{x} \quad (0 < y < x)$$

COMMON MISTAKE - AVOID THIS!

Common Mistakes:

1. Not specifying the range where $f_{X,Y}(x,y) > 0$
2. Forgetting to find marginal $f_Y(y)$ when computing conditional
3. Sign errors when computing CDFs from tail probabilities
4. Not checking that PDF integrates to 1

PROBLEM-SOLVING STRATEGY

PROBLEM-SOLVING STRATEGY:

Step 1: Identify what's given (marginal? conditional?)

Step 2: If computing expectation, try law of total expectation first

Step 3: If building joint PDF, use $f_{X,Y} = f_{Y|X} \cdot f_X$

Step 4: Always specify range! Draw picture if needed

Step 5: Sanity check: Does it integrate to 1? Makes sense?

4 EXPONENTIAL DISTRIBUTION & LIFETIMES

CRITICAL - MASTER THIS FIRST

Frequency: Appears in 5/7 exams (71%). Often combined with Poisson.

4.1 What is Exponential Distribution?

4.1.1 The Big Idea

Exponential distribution models **waiting times**:

- Time until next customer arrives
- Lifetime of a light bulb
- Time until your phone buzzes

Key parameter: λ = rate (events per unit time)

4.2 Key Facts

KEY FORMULAS

EXPONENTIAL DISTRIBUTION $T \sim \text{Exp}(\lambda)$:

PDF: $f(t) = \lambda e^{-\lambda t}$ for $t > 0$

CDF: $F(t) = 1 - e^{-\lambda t}$

Survival: $P(T > t) = e^{-\lambda t}$

Mean: $E[T] = \frac{1}{\lambda}$ (NOT $\lambda!$)

Variance: $\text{Var}(T) = \frac{1}{\lambda^2}$

COMMON MISTAKE - AVOID THIS!

Common Mistake: Confusing λ (rate) with $1/\lambda$ (mean)!

If rate is $\lambda = 2$ per minute, then mean time is $1/2 = 0.5$ minutes.

4.3 Memoryless Property

CRITICAL - MASTER THIS FIRST

The defining property of exponential distribution!

KEY FORMULAS

MEMORYLESS PROPERTY:

$$P(T > s + t \mid T > s) = P(T > t)$$

In words: If you've already waited s time, the probability of waiting t more is the same as from the start!

Key fact: Exponential is the ONLY continuous distribution with this property.

WORKED EXAMPLE

Problem: Bus arrives exponentially with mean 10 minutes. You've waited 5 minutes. What's the probability you wait 5 more minutes?

Solution:

$$P(T > 10 \mid T > 5) = P(T > 5) = e^{-5/10} = e^{-0.5} \approx \boxed{0.606}$$

The bus “doesn't remember” you've been waiting!

4.4 Minimum of Exponentials

KEY FORMULAS**MIN OF INDEPENDENT EXPONENTIALS:**

If $T_1 \sim \text{Exp}(\lambda_1)$ and $T_2 \sim \text{Exp}(\lambda_2)$ independent:

Distribution of min:

$$\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$$

Which happens first:

$$P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Expected time:

$$E[\min(T_1, T_2)] = \frac{1}{\lambda_1 + \lambda_2}$$

Connection: This is identical to competing Poisson processes!

4.5 Maximum of Exponentials

KEY FORMULAS**MAX OF INDEPENDENT EXPONENTIALS:**

Expected value:

$$E[\max(T_1, T_2)] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

Note: No simple form for the full distribution.

WORKED EXAMPLE

Problem: Bread making has two stages (happening simultaneously):

- Rising: mean 8 hours
- Proofing: mean 4 hours

What's the expected time until both are done?

Solution: Need $E[\max(T_1, T_2)]$ where $\lambda_1 = 1/8$, $\lambda_2 = 1/4$:

$$\begin{aligned} E[\max] &= \frac{1}{1/8} + \frac{1}{1/4} - \frac{1}{1/8 + 1/4} \\ &= 8 + 4 - \frac{1}{3/8} \\ &= 12 - \frac{8}{3} = 12 - 2.67 \\ &= \boxed{9.33 \text{ hours}} \end{aligned}$$

4.6 Exponential Quick Reference

KEY FORMULAS

EXPONENTIAL DISTRIBUTION - FORMULA SHEET

Basic Facts ($T \sim \text{Exp}(\lambda)$):

$$f(t) = \lambda e^{-\lambda t}$$

$$P(T > t) = e^{-\lambda t}$$

$$E[T] = \frac{1}{\lambda}$$

Memoryless: $P(T > s + t \mid T > s) = P(T > t)$

Min: $\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$

Max: $E[\max(T_1, T_2)] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$

5 BROWNIAN MOTION (Optional but Increasingly Common)

CRITICAL - MASTER THIS FIRST

Frequency: Appears in **3/7 exams (43%)**, but more common in recent years!

5.1 What is Brownian Motion?

Continuous random movement:

- Particle jiggling in water
 - Stock prices moving randomly
 - Position of drunk walker
- Like a random walk but in continuous time with infinitesimal steps.

5.2 Standard Brownian Motion

KEY FORMULAS

STANDARD BROWNIAN MOTION $B(t)$:

Definition:

1. $B(0) = 0$ (starts at origin)
2. $B(t) \sim N(0, t)$ (normally distributed)
3. Independent increments
4. $B(t) - B(s) \sim N(0, t - s)$ for $t > s$

Key Properties:

$$\begin{aligned} E[B(t)] &= 0 \\ \text{Var}(B(t)) &= t \\ E[B(t)^2] &= t \end{aligned}$$

Conditional:

$$\begin{aligned} E[B(t) \mid B(s) = x] &= x \quad (t > s) \\ E[B(t)^2 \mid B(s) = x] &= x^2 + (t - s) \quad (t > s) \end{aligned}$$

5.3 Geometric Brownian Motion (Stock Prices)

KEY FORMULAS

GEOMETRIC BROWNIAN MOTION:

$$S(t) = S_0 e^{B(t)}$$

Always positive (since $e^x > 0$), used to model stock prices.

WORKED EXAMPLE

Problem: Stock price $S(t) = e^{B(t)}$ starting at $S(0) = 1$.
What's $P(S(2) \geq S(1)^2)$?

Solution:

$$\begin{aligned} S(2) \geq S(1)^2 &\Leftrightarrow e^{B(2)} \geq (e^{B(1)})^2 \\ &\Leftrightarrow e^{B(2)} \geq e^{2B(1)} \\ &\Leftrightarrow B(2) \geq 2B(1) \\ &\Leftrightarrow B(2) - 2B(1) \geq 0 \end{aligned}$$

Now find distribution of $B(2) - 2B(1)$:

$$B(2) - 2B(1) = [B(2) - B(1)] - B(1)$$

where $B(2) - B(1) \sim N(0, 1)$ and $B(1) \sim N(0, 1)$ are independent.
Therefore $B(2) - 2B(1) \sim N(0, 1 + 1) = N(0, 2)$.

$$\begin{aligned} P(B(2) - 2B(1) \geq 0) &= P\left(\frac{B(2) - 2B(1)}{\sqrt{2}} \geq 0\right) \\ &= P(Z \geq 0) \quad \text{where } Z \sim N(0, 1) \\ &= \boxed{0.5} \end{aligned}$$

6 PATTERN RECOGNITION GUIDE

CRITICAL - MASTER THIS FIRST

The Secret to Success: Recognize the problem type instantly, apply the right technique!

6.1 When You See... Use This!

Keywords/Setup	Technique to Use
“Poisson process” + “types/categories”	THINNING: Type A rate = λp
“which happens first” + rates	COMPETING PROCESSES: $P(A \text{ first}) = \lambda_A / (\lambda_A + \lambda_B)$
“given that n events occurred”	CONDITIONAL EXPECTATION: $E[T N = n] = n/\lambda$
“random walk” + “probability of reaching”	FIRST-STEP ANALYSIS for hitting probability $p(i)$
“random walk” + “expected time/steps”	FIRST-STEP ANALYSIS for expected time $\mu(i)$ (DON'T FORGET +1!)
“joint density” or “ $f(x, y)$ ”	Set up JOINT PDF , use $f_{X,Y} = f_{Y X} \cdot f_X$
“given $X = x$, find $E[Y]$ ”	Use LAW OF TOTAL EXPECTATION: $E[Y] = E[E[Y X]]$
“ $Y = UX$ ” where $U \sim \text{Unif}[0, 1]$	$f_{Y X}(y x) = 1/x$ for $0 < y < x$
“exponential” + “min” or “first”	MIN OF EXPONENTIALS: $\min \sim \text{Exp}(\lambda_1 + \lambda_2)$
“exponential” + “max” or “both done”	$E[\max] = 1/\lambda_1 + 1/\lambda_2 - 1/(\lambda_1 + \lambda_2)$
“Brownian motion” or “ $B(t)$ ”	Use BM properties: $E[B(t)] = 0$, $\text{Var}(B(t)) = t$
“ $S(t) = e^{B(t)}$ ” (stock prices)	GEOMETRIC BM: Convert to statements about $B(t)$

6.2 Problem Type Flowchart

- Does it mention “arrivals” or “rate”?
 - YES → Probably Poisson process
 - Check for types/categories → Use thinning
 - Check for “which first” → Competing processes
- Does it mention “positions”, “game”, “walk”?
 - YES → Random walk with first-step analysis
 - Asked for probability? → Set up $p(i)$ equations
 - Asked for expected time? → Set up $\mu(i)$ equations
- Does it involve **TWO** random variables?
 - YES → Joint/conditional densities
 - Check for $Y = UX$ → Use standard formulas
 - Need expectation? → Try law of total expectation
- Does it mention “waiting time” or “lifetime”?
 - YES → Probably exponential
 - Multiple waiting times? → Min/max of exponentials
- Does it mention “Brownian motion” or $B(t)$?
 - YES → Use BM properties
 - Stock prices? → Geometric BM

7 COMPLETE FORMULA SHEET

7.1 Poisson Processes

Basic: $N(t) \sim \text{Poisson}(\lambda t)$, $E[N(t)] = \text{Var}(N(t)) = \lambda t$

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

Thinning: Type A prob $p \rightarrow$ Type A is $\text{Poisson}(\lambda p)$

Competing: $P(A \text{ before } B) = \frac{\lambda_A}{\lambda_A + \lambda_B}$, $E[T] = \frac{1}{\lambda_A + \lambda_B}$

Conditional: $E[T \mid N(T) = n] = \frac{n}{\lambda}$

7.2 Random Walks

Hitting Probability:

$$p(i) = p_{\uparrow}p(i+1) + p_{\downarrow}p(i-1) + p_{\leftrightarrow}p(i)$$

Boundaries: $p(a) = 0$, $p(b) = 1$

Expected Time:

$$\mu(i) = 1 + p_{\uparrow}\mu(i+1) + p_{\downarrow}\mu(i-1) + p_{\leftrightarrow}\mu(i)$$

Boundaries: $\mu(a) = 0$, $\mu(b) = 0$

Symmetric Shortcuts ($p_{\uparrow} = p_{\downarrow} = 1/2$):

$$p(i) = \frac{i-a}{b-a}, \quad \mu(i) = (i-a)(b-i)$$

7.3 Joint & Conditional Densities

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}, \quad f_{X,Y}(x,y) = f_{Y|X}(y|x) \cdot f_X(x)$$

$$f_X(x) = \int f_{X,Y}(x,y) dy$$

Law of Total Expectation: $E[Y] = E[E[Y|X]]$

For $Y = UX$, $U \sim \text{Unif}[0,1]$:

$$f_{Y|X}(y|x) = \frac{1}{x} \text{ for } 0 < y < x, \quad E[Y|X=x] = \frac{x}{2}$$

7.4 Exponential Distribution

$T \sim \text{Exp}(\lambda)$: $f(t) = \lambda e^{-\lambda t}$, $P(T > t) = e^{-\lambda t}$, $E[T] = \frac{1}{\lambda}$

Memoryless: $P(T > s+t \mid T > s) = P(T > t)$

Min: $\min(T_1, T_2) \sim \text{Exp}(\lambda_1 + \lambda_2)$, $P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$

Max: $E[\max(T_1, T_2)] = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$

7.5 Brownian Motion

$B(t)$ **standard BM:** $E[B(t)] = 0$, $\text{Var}(B(t)) = t$, $B(t) \sim N(0, t)$

$B(t) - B(s) \sim N(0, t - s)$ for $t > s$

Conditional: $E[B(t)^2 \mid B(s) = x] = x^2 + (t - s)$ for $t > s$

Geometric BM: $S(t) = S_0 e^{B(t)}$ (stock prices)

8 COMMON MISTAKES & HOW TO AVOID THEM

8.1 Top 10 Mistakes Students Make

1. **Forgetting “+1” in random walk expected time**
 - Formula: $\mu(i) = 1 + \dots$ (THE 1 IS CRITICAL!)
 - Fix: Write it in red on your cheat sheet
2. **Not using thinning when events split into types**
 - Keyword: “40% order coffee, 30% tea...”
 - Fix: Always check if arrivals split by category
3. **Confusing λ with $1/\lambda$ for exponential**
 - Rate $\lambda = 2/\text{min}$ means mean time $= 1/2 = 0.5$ min
 - Fix: Write “ $E[T] = 1/\lambda$ ” on cheat sheet
4. **Wrong boundary conditions for random walks**
 - For $p(i)$: $p(a) = 0, p(b) = 1$
 - For $\mu(i)$: $\mu(a) = 0, \mu(b) = 0$
 - Fix: Always write boundaries first
5. **Not specifying range for joint PDFs**
 - Must say “for $0 < y < x$ ” or similar
 - Fix: Draw a picture of the region
6. **Not checking if random walk is symmetric**
 - If $p_{\uparrow} = p_{\downarrow} = 1/2$, use shortcuts!
 - Fix: First check symmetry before setting up equations
7. **Arithmetic errors in solving systems**
 - First-step analysis needs solving 2-3 equations
 - Fix: Show all work, check answer makes sense
8. **Using λt instead of λ in Poisson**
 - Rate is λ , expected count in time t is λt
 - Fix: Be clear about units (per minute vs total)
9. **Forgetting independence assumptions**
 - Thinning requires independence
 - Min of exponentials requires independence
 - Fix: Check problem statement
10. **Not showing work / skipping steps**
 - Partial credit is huge!
 - Fix: Write every step, even if obvious

9 EXAM DAY STRATEGY

9.1 Before the Exam

1. **Sleep well** - Seriously, don't pull an all-nighter
2. **Prepare cheat sheet** - One page, both sides, handwritten
3. **Eat something** - Low energy = mistakes
4. **Arrive early** - Get settled, calm down

9.2 During the Exam

PROBLEM-SOLVING STRATEGY

FIRST 5 MINUTES:

1. Skim all questions quickly
2. Identify problem types (Poisson? Random walk? Joint density?)
3. Star the ones you feel confident about
4. Budget time: 35 minutes per question

PROBLEM-SOLVING STRATEGY

FOR EACH PROBLEM:

1. **Read carefully** - What's given? What's asked?
2. **Identify technique** - Use pattern recognition guide
3. **Write what you know** - Definitions, parameters, distributions
4. **Show all work** - Partial credit!
5. **Check units** - Per minute? Per hour? Total?
6. **Simplify answer** - Reduce fractions
7. **Sanity check** - Does it make sense?

9.3 Time Management

- **Don't get stuck** - If blocked after 5 min, move on
- **Do easy parts first** - Get partial credit
- **Come back to hard parts** - Fresh perspective helps
- **Leave 5 minutes** - Final check, upload

9.4 Partial Credit Strategy

Even if you can't solve fully:

- Write down relevant formulas
- Set up the problem (define variables)
- Explain your approach in words
- Show partial calculations
- Write "If I could solve X, then I would do Y"

10 STUDY PLAN

10.1 4-Day Intensive Plan

10.1.1 Day 1: Poisson Processes (8 hours)

Morning (4 hours):

- Read Section 1 of this guide (2 hours)
- Work through all examples yourself (2 hours)

Afternoon (4 hours):

- Practice: F2025 Q3 (Coffee shop)
- Practice: S2025 Q3 (Taxi stop)
- Practice: 2019 Q1 (Plagues)
- Check solutions carefully

Evening Check:

- Can you explain thinning without notes?
- Can you solve competing process problems?

10.1.2 Day 2: Random Walks (8 hours)

Morning (4 hours):

- Read Section 2 of this guide (2 hours)
- Practice writing first-step equations (2 hours)

Afternoon (4 hours):

- Practice: F2025 Q2 (Sleepy gambler)
- Practice: S2025 Q1 (Flight prices)
- Practice: 2017 Q3 (Asymmetric game)

Evening Check:

- Write formulas from memory
- Do you remember the “+1”?

10.1.3 Day 3: Joint Densities + Full Exam (8 hours)

Morning (3 hours):

- Read Section 3 (2 hours)
- Practice: F2025 Q1 (Insurance) (1 hour)

Afternoon (5 hours):

- Full timed practice exam: F2025 or S2025 (2 hours)
- Check solutions thoroughly (2 hours)
- Quick review of exponential & BM (1 hour)

10.1.4 Day 4: Review + Weak Spots (6 hours)

Morning (3 hours):

- Another full practice exam
- Identify your 2-3 weakest topics

Afternoon (3 hours):

- Redo problems you got wrong
- Focus on weak topics only
- Make cheat sheet

Evening:

- Light review of formula sheet
- Get good sleep!

10.2 2-Day Crash Plan

If you only have 2 days:

10.2.1 Day 1: Poisson + Random Walks (12 hours)

- These appear in 100% of exams
- Read Sections 1 & 2
- Do 3-4 practice problems each
- Master thinning and first-step analysis

10.2.2 Day 2: Everything Else + Practice (12 hours)

- Skim Section 3 (joint densities)
- Do 2 full practice exams
- Focus on understanding solutions
 - **Skip:** Brownian motion if pressed for time

11 FINAL THOUGHTS

11.1 You Can Do This!

Remember:

- This exam is **predictable** - same topics every year
- Master **3 techniques** = handle 75% of exam
- Understanding > memorizing
- Partial credit is huge
- Thousands have learned this before you

11.2 The Formula for Success

Success = Pattern Recognition + Practice + Showing Work

- **Pattern Recognition:** See “types” → Think thinning
- **Practice:** Do problems, check solutions, understand mistakes
- **Show Work:** Write every step = maximum partial credit

11.3 What to Focus On

Topic	Time Investment
Poisson Processes (esp. thinning)	30%
Random Walks (first-step analysis)	30%
Joint/Conditional Densities	20%
Exponential Distribution	10%
Practice Exams	10%

11.4 Last-Minute Checklist

Before Exam:

- ☐ Cheat sheet prepared (formulas, NOT examples)
- ☐ Calculator ready (if allowed)
- ☐ Reviewed formula sheet (Section 8)
- ☐ Did at least 2 full practice exams
- ☐ Got good sleep

During Exam:

- ☐ Read all questions first
- ☐ Budget 35 minutes per question
- ☐ Show all work (partial credit!)
- ☐ Check answers make sense
- ☐ Upload with 5 minutes to spare

Good luck!

You’ve got the tools. You’ve got the guide. Now go ace this exam!

“Success is not final, failure is not fatal: it is the courage to continue that counts.”
— Winston Churchill

Master these concepts, ace the exam!